

Confusion matrix?

A **confusion matrix** is a performance measurement tool for machine learning classification problems.

It shows how well a model's predictions match the actual classes.

It is especially useful for evaluating **binary** and **multi-class classification** models.

◆ Confusion Matrix Structure (Binary Classification)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

✓ Explanation of Each Term

- TP (True Positive): Model correctly predicted the positive class.
- TN (True Negative): Model correctly predicted the negative class.
- FP (False Positive / Type I Error): Model incorrectly predicted positive when it's actually negative.
- FN (False Negative / Type II Error): Model incorrectly predicted negative when it's actually positive.



Metrics Derived from Confusion Matrix

1. **Accuracy** = $(TP + TN) / (TP + TN + FP + FN)$

Overall correctness of the model.

2. **Precision** = $TP / (TP + FP)$

How many predicted positives are actually positive?

3. **Recall (Sensitivity)** = $TP / (TP + FN)$

How many actual positives are correctly predicted?

4. **F1 Score** = $2 \times (Precision \times Recall) / (Precision + Recall)$

Harmonic mean of Precision and Recall.



Example

Let's say we have:

- $TP = 70$
- $TN = 50$
- $FP = 10$
- $FN = 20$

Then:

- $Accuracy = (70+50)/(70+50+10+20) = 120/150 = 0.8$
- $Precision = 70 / (70+10) = 0.875$
- $Recall = 70 / (70+20) = 0.778$
- $F1\ Score = 2 \times (0.875 \times 0.778) / (0.875 + 0.778) \approx 0.823$

